

MAC Address Table :

A switch stores MAC addresses of connected devices in a MAC Address Table.

Older Name : CAM Table

Purpose :

The MAC Address Table maps
MAC Address $\rightarrow$ switch port

MAC Learning :

Initially,

The MAC Address Table is empty.
when a device sends a frame :

- 1) The switch reads the source
MAC address.
- 2) It learns which port that MAC
address belongs to.
- 3) It store the entry in the MAC
address table.

Frame Forwarding :

After learning MAC addresses.

The switch forwards frames only to the destination port

unlike a hub,

The switch does not flood every frame to all ports.

Results :

- Less unnecessary traffic
- Better bandwidth utilization
- Fewer collisions
- Higher throughput

Unknown Destination MAC :

If the destination MAC address is not in the MAC address table,

The switch floods the frame to all ports except the incoming port.

Once the destination replies,

The switch learns its MAC address